

Garvit Singh Rathore

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Backend-focused Software Engineering Student building scalable distributed systems, low-latency APIs, and fault-tolerant cloud services with Java, Spring Boot, AWS, and Redis.

Education

SRM Institute of Science and Technology
B.Tech in Computer Science Engineering

Chennai, India
2024 – 2028

Experience

Software Engineering Intern – FOSSEE, IIT Bombay
Open Steel Design and Graphics

Oct 2025 – Present
Mumbai/Remote

- Contributed 10,000+ lines of production-grade code to Osdag, an open-source engineering platform used for structural design workflows.
- Refactored core computation modules, reducing redundant processing by 30% and improving execution efficiency across large design workloads.
- Resolved GUI-to-business-logic inconsistencies and implemented schema-driven UI rendering, improving reliability and maintainability.
- Collaborated in an open-source distributed environment using Git, code reviews, and issue-driven development practices.

Open Source Contributor
ArduPilot & Osdag

2021 – Present
Remote

- Contributed to modular real-time systems involving CAN communication, reliability improvements, and fault-tolerant software architecture.

Undergraduate Researcher – UROP
SRM Institute of Science and Technology

2025 – Present
Chennai

- Built machine learning pipelines for neuroimaging signal analysis focused on early dyslexia detection.
- Designed optimized feature extraction workflows improving classification accuracy by 15%.

Projects

Enterprise Transaction API (Java, Spring Boot, PostgreSQL)

2025

- Built high-performance REST APIs handling 5,000+ concurrent requests with ACID-compliant transaction guarantees.
- Reduced average API latency by 120ms using caching, optimized SQL queries, and connection pooling strategies.
- Implemented secure OAuth2 authentication, RBAC authorization, and robust exception handling for sensitive financial-style workflows.
- Designed service layers with transactional consistency and rollback support for multi-step operations.

Distributed Task Queue System (Python, Redis)

2026

- Engineered a distributed asynchronous task queue using Redis supporting execution across 5 worker nodes.
- Implemented retry logic, dead-letter handling, worker recovery, and queue scheduling for 99.9% job reliability.
- Improved throughput by 40% through connection pooling and optimized producer-consumer architecture.

Cloud-Native Backend Service (Java, AWS, Microservices)

2024

- Designed microservices-based backend systems on AWS for scalable inter-service communication and resilient deployment.
- Implemented fault isolation, service recovery, and secure data serialization mechanisms.
- Focused on scalability, observability, and high availability patterns aligned with enterprise backend systems.

Technical Skills

Languages: Java, Python, C/C++, JavaScript, TypeScript, SQL

Backend: Spring Boot, REST APIs, Microservices, Node.js, Redis, System Design

Databases: PostgreSQL, MySQL, MongoDB, DynamoDB

Cloud & DevOps: AWS (EC2, S3, Lambda, VPC), Docker, Linux, CI/CD

Core CS: Data Structures, Algorithms, OOP, Operating Systems, DBMS, Distributed Systems, Computer Networks

Achievements & Certifications

- Solved 500+ algorithmic problems on LeetCode across trees, graphs, DP, greedy, and system-design-oriented problems.
- AWS Certified Solutions Architect – Associate (Ongoing)
- NPTEL (SWAYAM): Data Structures & Algorithms, Object Oriented Programming, Programming in Java